

Corrigé — Examen d'entraînement LEPL1110 — Éléments Finis

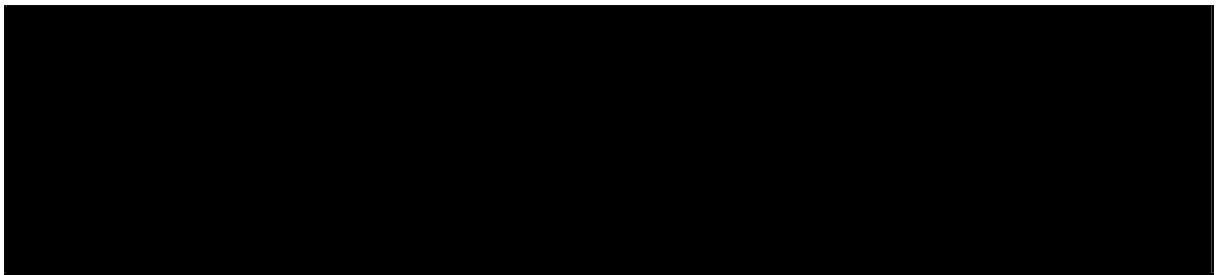
Claude code

Mai 2026

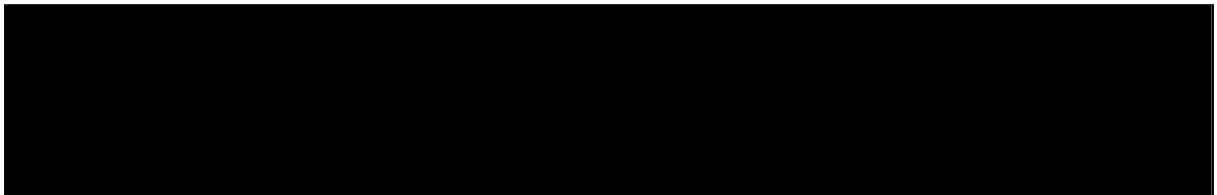
Exercice 1 — Contraintes et méthodes de résolution

1.1 — Problème sans contrainte et formulation du problème contraint

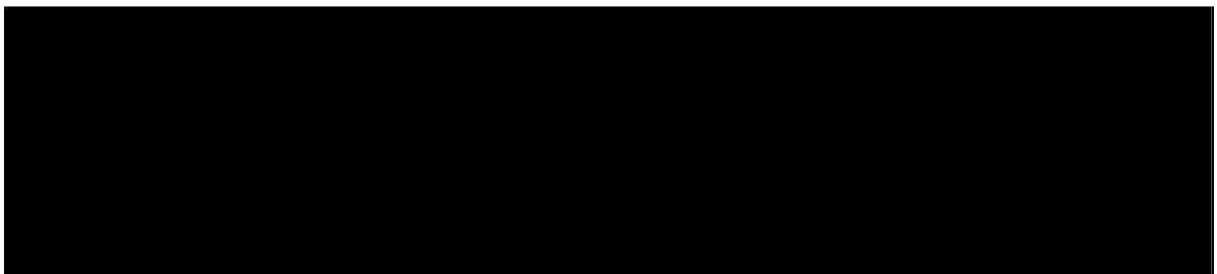
(a)



(b)



(c)

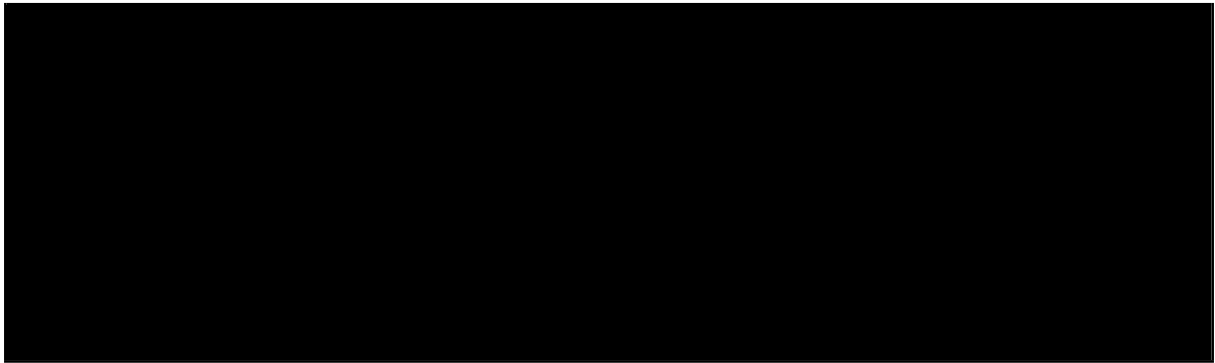


1.2 — Méthode des multiplicateurs de Lagrange

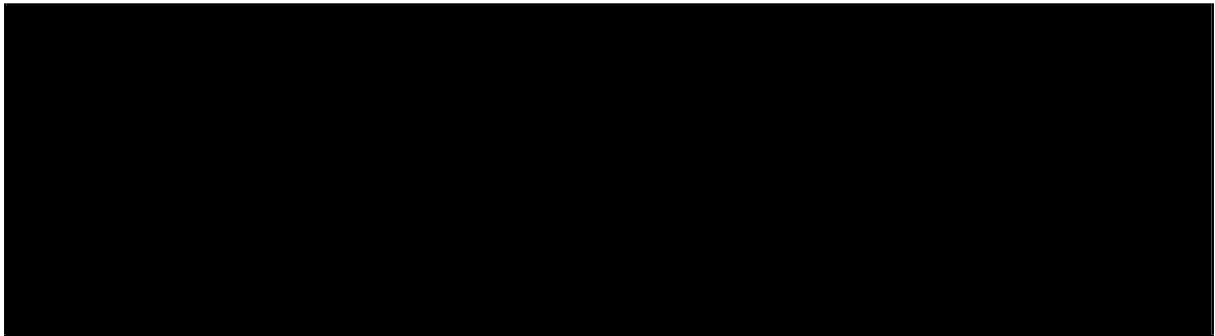
(a)



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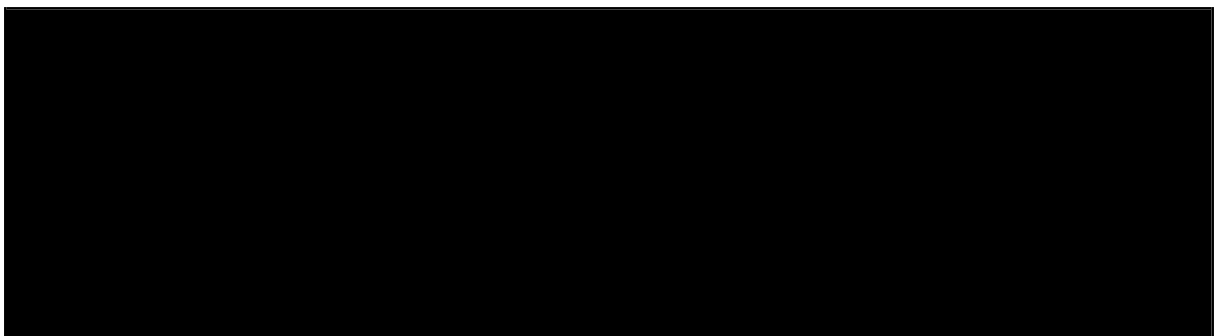


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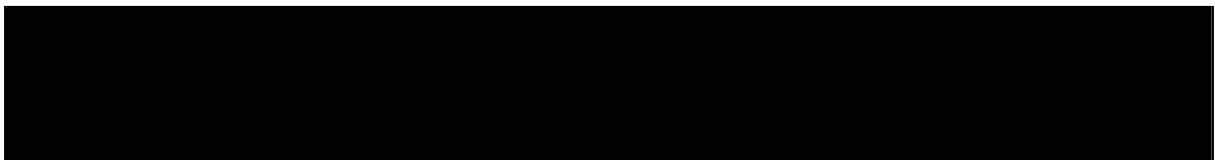


1.3 — Méthode de pénalité

(a)



(b)



(c)

1.4 — Comparaison des deux méthodes

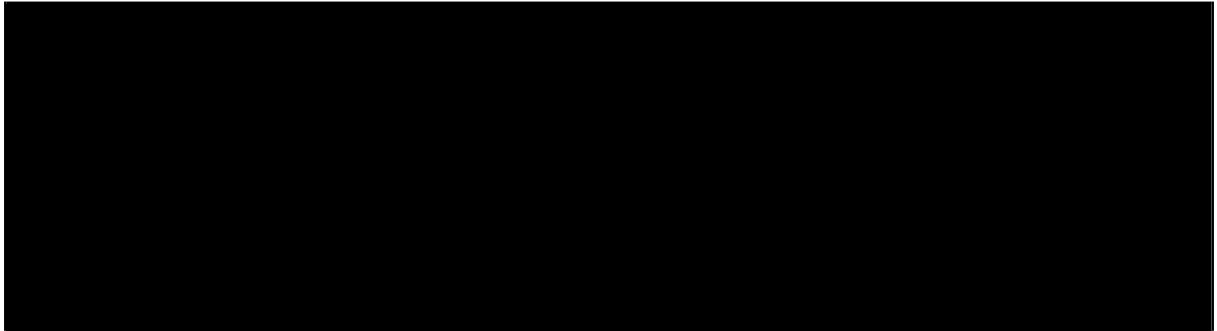
Exercice 2 — Stabilité et précision des éléments mixtes

2.1 — Formulation faible mixte

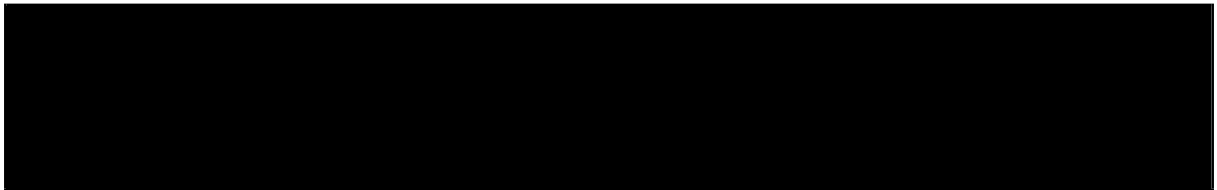
(a)



(b)

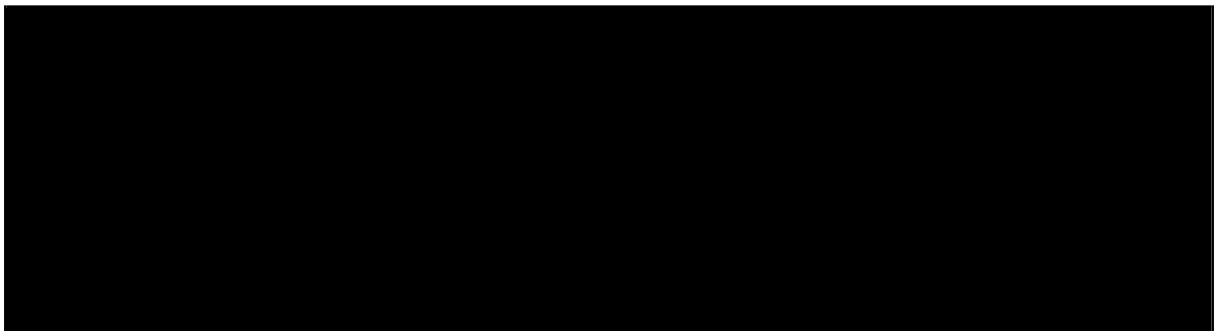


(c)

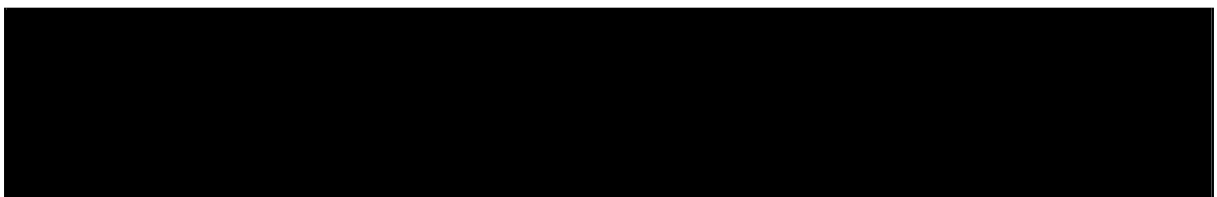


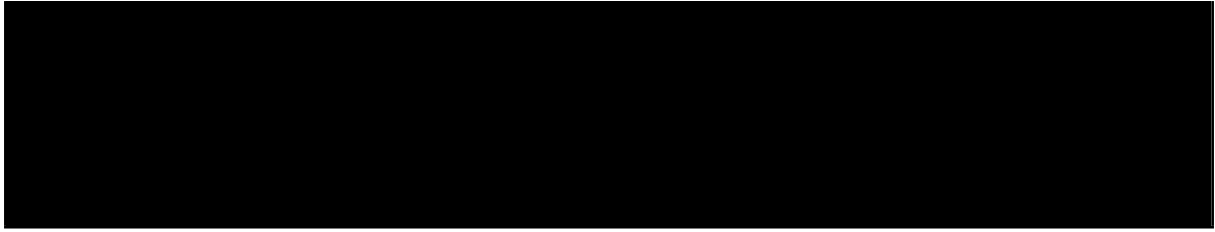
2.2 — Condition LBB

(a)

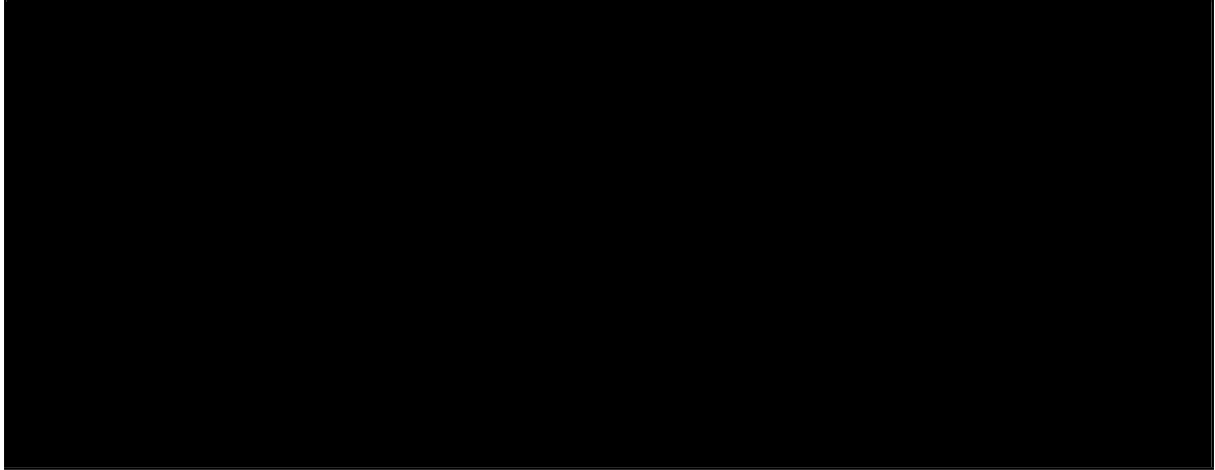


(b)



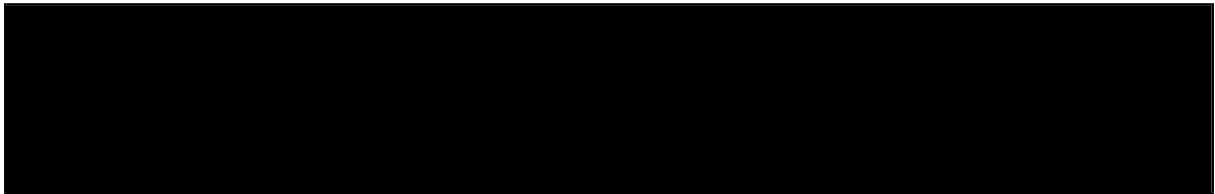


(c)

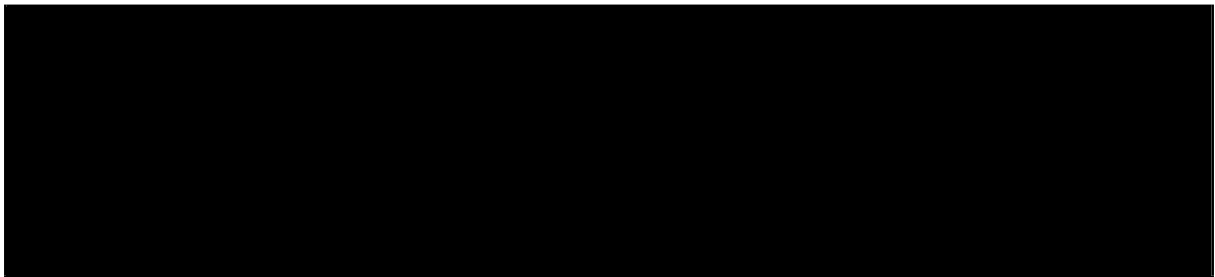


2.3 — Estimation de précision

(a)

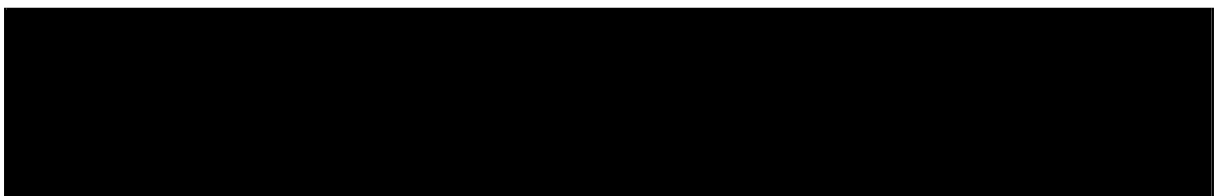


(b)



2.4 — Classification des éléments mixtes

(a)

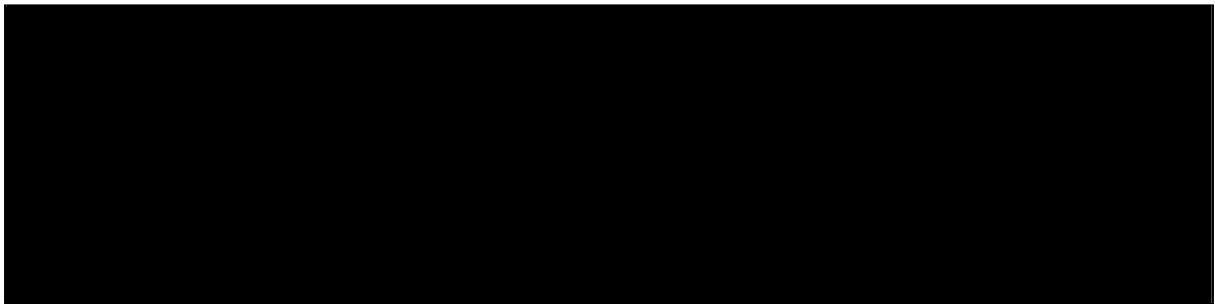




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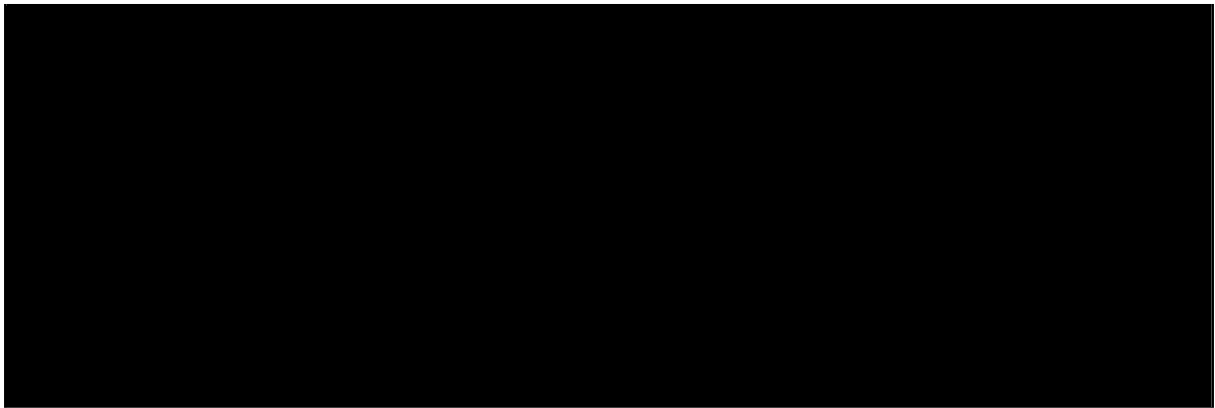


(c)



2.5 — Pression continue vs discontinue

(a)



(b)

